MATH 2211, SPRING 2026: HOMEWORK 3

(1) Decide if the following statements are true or false. Give a line or two in justification for your answer. It is permissible for that justification to be a reference to previous homeworks or to lectures:
  (a) $\mathbb{Z}/4\mathbb{Z}$ is a field under modular arithmetic.
  (b) If F is a field and $a, b \in F$ are such that $a \cdot b = 0$, then one of a or b must be 0.
  (c) The set $\mathrm{Poly}_3(F)$ of polynomials of degree 3 with coefficients of F is a subspace of $\mathrm{Poly}(F)$.
  (d) The equation $z^n = a$ for $a \in \mathbb{C}$, $a \neq 0$, admits n distinct solutions in $\mathbb{C}$.
  (e) There exists a complex number z such that $z\overline{z} - 1 = i$.

(2) Decide if the following subsets are $\mathbb{R}$-vector subspaces. Again, justify your responses.[1]
  (a) $U_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : \; x_1 + 2x_2 + 3x_3 = 0\} \subset \mathbb{R}^3$;
  (b) $U_2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : \; x_1 x_2 x_3 = 0\} \subset \mathbb{R}^3$;
  (c) $U_3 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : \; x_1 + 2x_2 + 3x_3 = 1\} \subset \mathbb{R}^3$;
  (d) $U_4 = \{f(x) \in \mathrm{Poly}(\mathbb{R}) : \; f'(-1) = 3f(2)\} \subset \mathrm{Poly}(\mathbb{R})$;
  (e) $U_5 = \{A \in M_{2\times 2}(\mathbb{R}) : \; \det(A) = 0\} \subset M_{2\times 2}(\mathbb{R})$;

  Here, $M_{2\times 2}(\mathbb{R})$ is the $\mathbb{R}$-vector space of 2×2-matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\det(A)$ is the *determinant* $ad - bc$.

(3) Let V be an F-vector space.
  (a) If $U_1, U_2 \subset V$ are subspaces, show that the subsets
$$U_1 + U_2 = \{u_1 + u_2 : \; u_1 \in U_1, u_2 \in U_2\} \subset V \; ; \; U_1 \cap U_2 \subset V$$
  are also subspaces.
  (b) Prove or give a counterexample: If $U_1, U_2, W \subset V$ are all subspaces, and if $U_1 + W = U_2 + W$, then $U_1 = U_2$.

(4) Is $w = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \in \mathbb{C}^3$ a linear combination of the following vectors?
$$\begin{bmatrix} 1 \\ 1 \\ -i \end{bmatrix}, \; \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \; \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}.$$

(5) Is the polynomial $x + 1 \in \mathrm{Poly}(\mathbb{R})$ a linear combination of the following polynomials?
$$x^2 + 1, x^3 + x, 2x^2 + x, x + 3 \in \mathrm{Poly}(\mathbb{R}).$$

[1]We will use 'tuple' notation instead of column vector notation to save space.